Cut Rules in DAG-Based BFT Mempools: Comparing Relaxed Mempool and Autobahn

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1. Motivation: The Cut Rule Problem

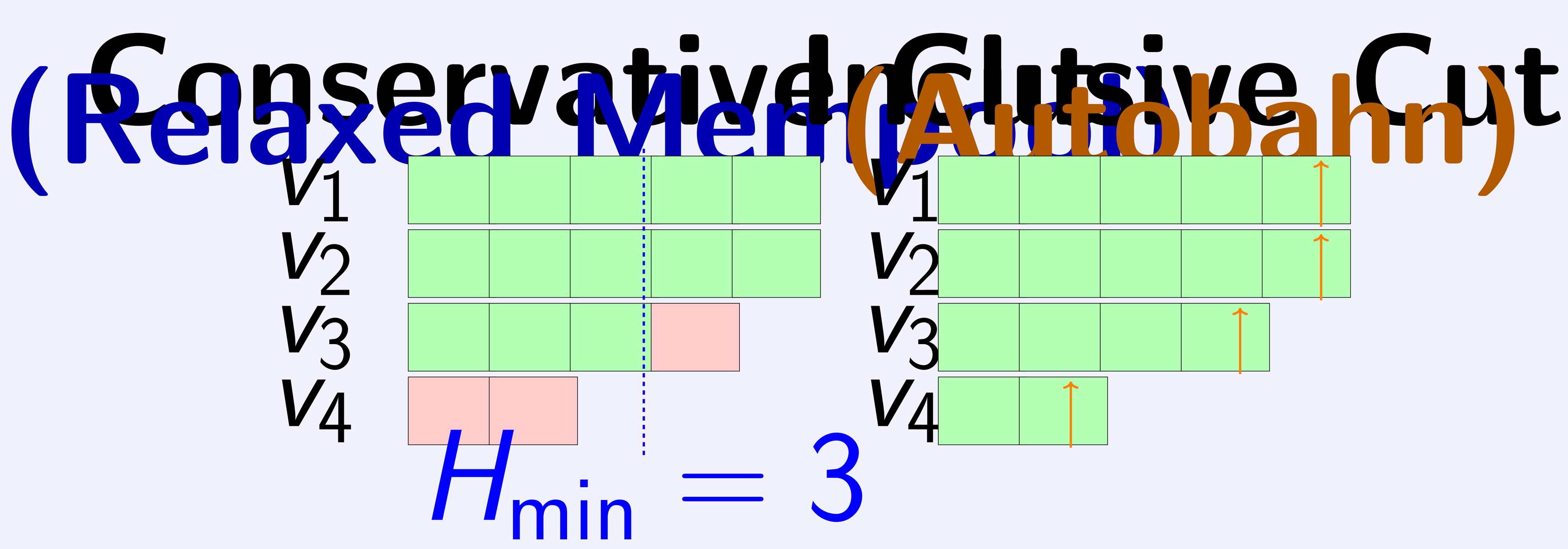
Modern DAG-based BFT protocols decouple data dissemination from ordering. Validators broadcast batches into a DAG; a consensus leader periodically **cuts** the DAG to commit transactions.

Key Question: How should the leader decide *which* data to include in each commit?

This choice—the **cut rule**—determines:

- **Straggler resilience:** Do slow validators delay commits?
- **Transaction fairness:** Are all validators' data eventually included?
- **Hangover effect:** How fast does the system recover after blips?

2. Two Cut Strategies



Conservative (RM):
Top $2f+1$ by height
Uniform H_{min} cut
Slow nodes excluded
Sync before finalization

Inclusive (Autobahn):
All n lanes included
Per-lane tip references
No node excluded
Sync after voting

3. Key Insight

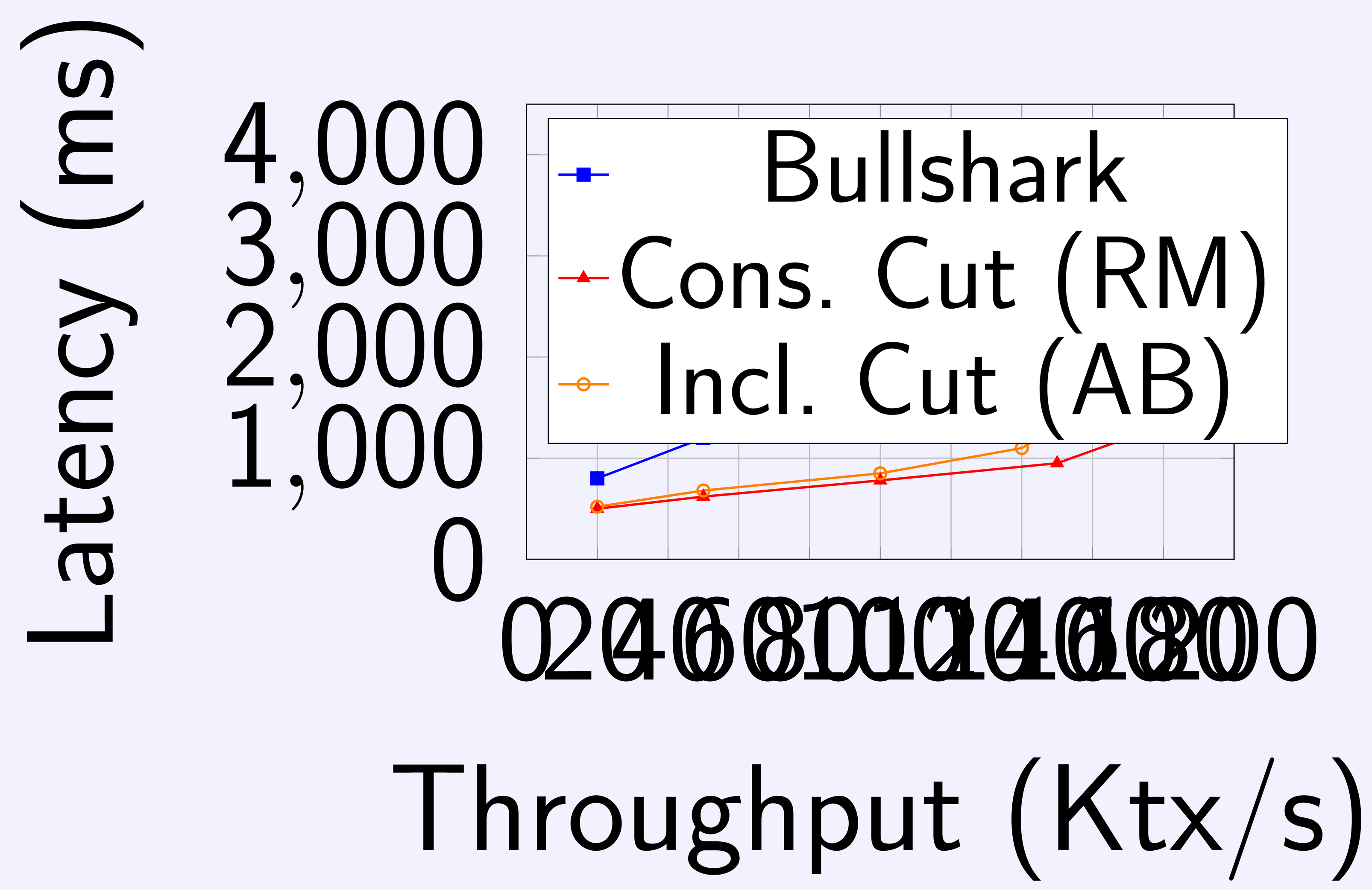
Both systems share the **same broadcast quorum** ($f+1$) for speculative data availability, but diverge in how the leader constructs commit proposals:

Property	Relaxed Mempool	Autobahn
Cut scope	Top $2f+1$	All n lanes
Slow node policy	Exclude	Include at own tip
Commit size	$O(H_{min} \cdot n)$	$O(n)$ tip refs
Hangover	Gradual recovery	Instant recovery

4. Experimental Results

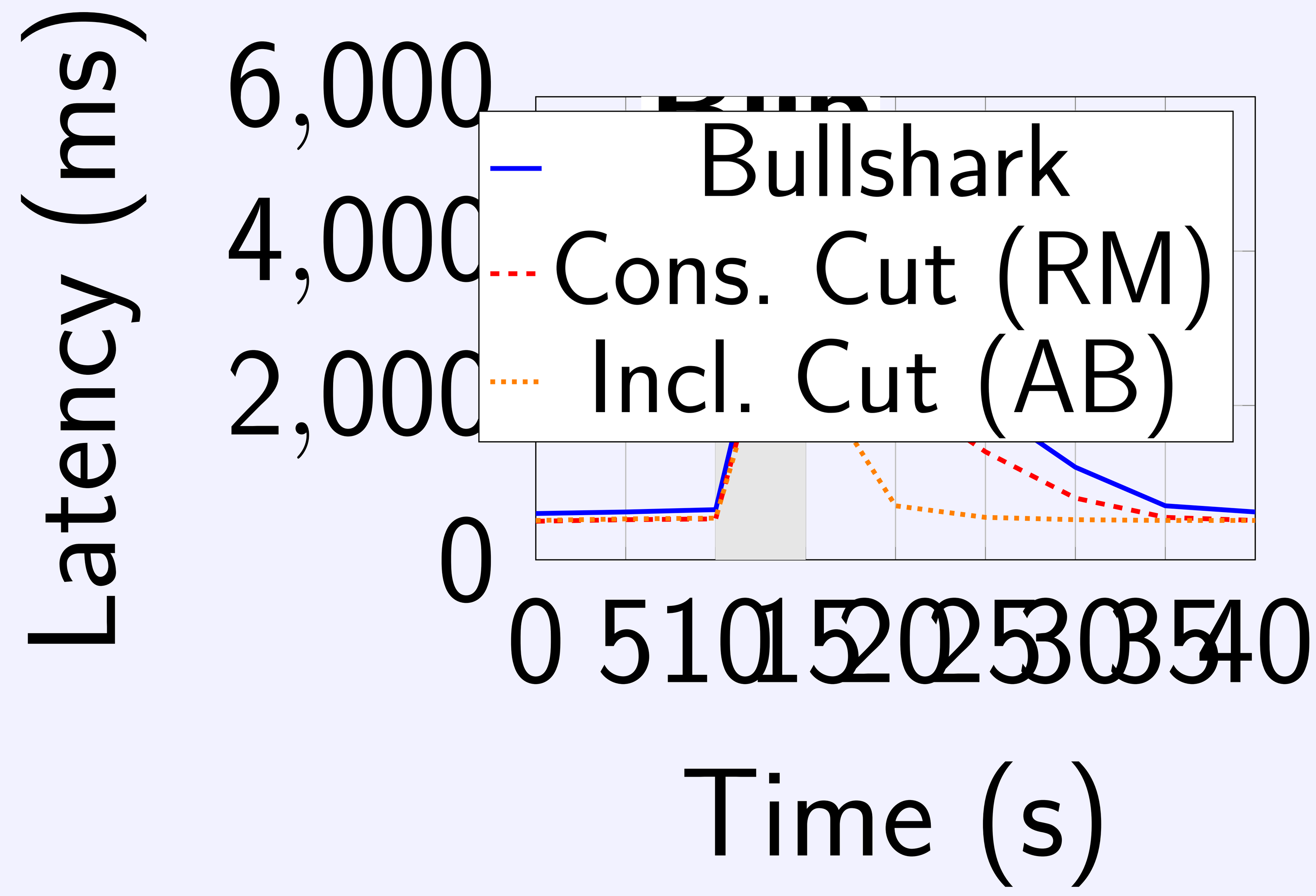
Setup: 4 servers (i9-13900K, 64 GB), 10 validators, 512 B tx, 500 KB batches.

(a) Straggler Scenario (4 stragglers, +300 ms delay)



Conservative cut: **170K tx/s at 1.2s** (excludes slow nodes).
Inclusive cut: 160K tx/s at 1.5s (includes all, sync overhead).
Bullshark: 98K tx/s at 4s (round-based, waits for all).

(b) Hangover Effect (5s network blip at t=10s)



Inclusive cut recovers in ~**2s** (one slot commits all backlog).
Conservative cut: ~20s (H_{min} advances gradually).
Bullshark: ~25s (sequential round processing).

5. Conclusion & Future Work

Neither cut strategy dominates universally:

- **Conservative cut:** Best for persistent stragglers (lower latency)
- **Inclusive cut:** Best for blip-prone environments (seamless recovery)

Future Direction: *Adaptive cut* that monitors validator height disparity and dynamically switches between conservative and inclusive modes. Both strategies share the same DAG substrate, making runtime switching feasible.

References

[1] Danezis et al., "Narwhal and Tusk," *EuroSys*, 2022.
[2] Spiegelman et al., "Bullshark," *ACM CCS*, 2022.
[3] Giridharan et al., "Autobahn: Seamless high speed BFT," *SOSP*, 2024.